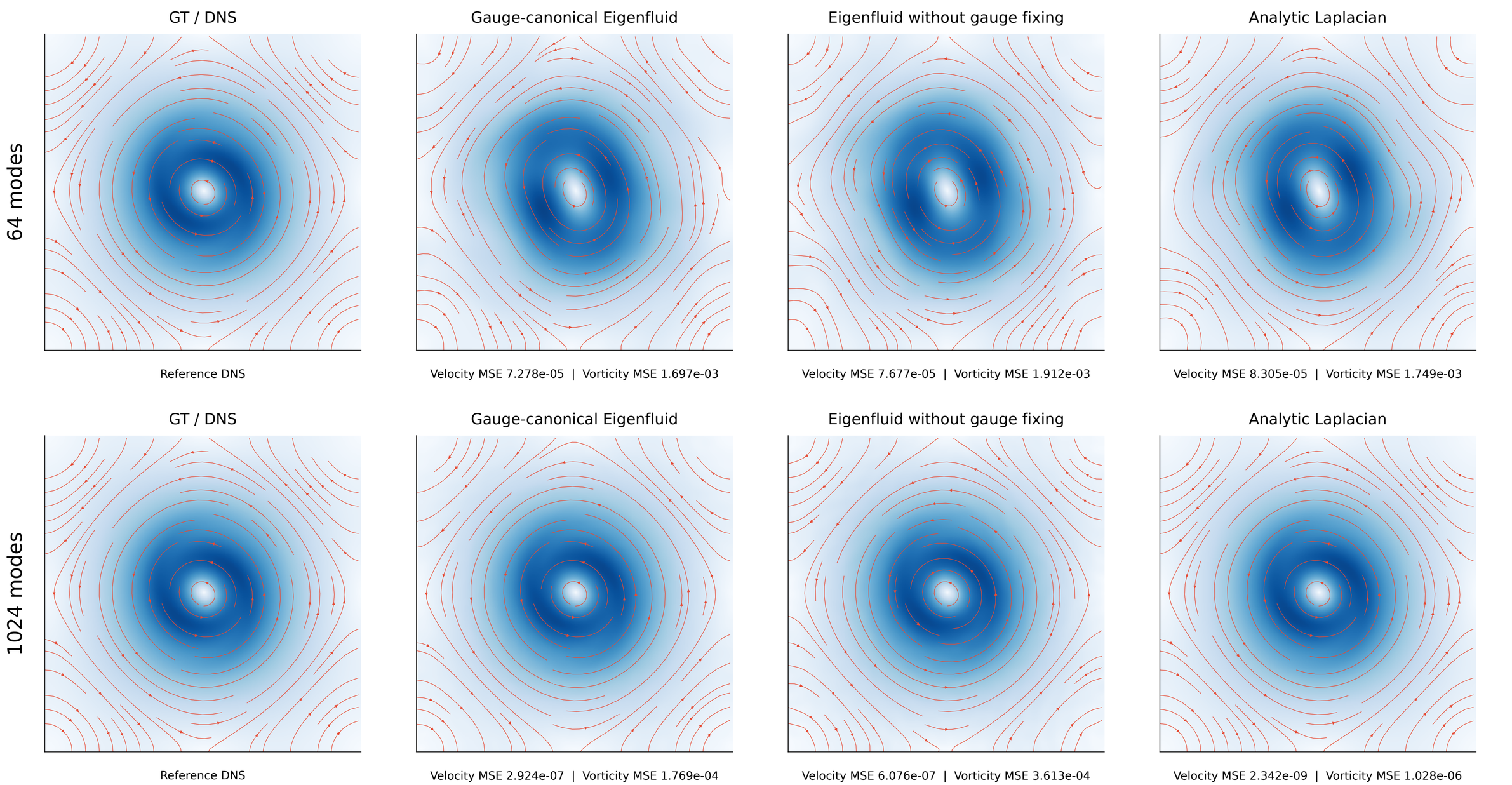

Vortex merger: velocity and streamlines | t = 18.8496 | frame 61/61
Re = 350 | vortex separation = 0.90 | 128 x 128 evaluation grid | identical color scale across all panels
GT / DNS
Gauge-canonical Eigenfluid
Eigenfluid without gauge fixing
Analytic Laplacian
64 modes
Reference DNS
Velocity MSE 7.278e-05 | Vorticity MSE 1.697e-03
Velocity MSE 7.677e-05 | Vorticity MSE 1.912e-03
Velocity MSE 8.305e-05 | Vorticity MSE 1.749e-03
GT / DNS
Gauge-canonical Eigenfluid
Eigenfluid without gauge fixing
Analytic Laplacian
1024 modes
Reference DNS
Velocity MSE 2.924e-07 | Vorticity MSE 1.769e-04
Velocity MSE 6.076e-07 | Vorticity MSE 3.613e-04
Velocity MSE 2.342e-09 | Vorticity MSE 1.028e-06
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Same GT frame 0 projected into each basis; autonomous Navier-Stokes Galerkin rollout thereafter. No later GT refitting.